
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations, but bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis. In vision, we fully reproduce the core findings: weight decay reduces effective rank from 38.5 to 21.4, confirming that regularization induces interpretable low-rank structure, while eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy. In language, we confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” form opposing directions (cosine similarity -0.16). However, we do *not* reproduce the low-rank interaction claim: the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%, and the mean rank-2 correlation saturates at 0.37–0.55. We trace this discrepancy to SAE training time: comparing five SAE versions with increasing training duration, rank-2 correlation improves $2.5\times$ ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that correlation depends on SAE quality. We provide evidence that the publicly available SAE artifacts have not reached the convergence levels of the internal checkpoints used in the original paper, creating a reproducibility gap with practical implications: while the bilinear decomposition methodology is sound, low-rank approximation quality depends critically on SAE convergence—a factor warranting explicit attention in future applications. In our extensions, we demonstrate that regularized bilinear MLPs learn transferable geometric circuits—MNIST-trained models achieve 97.7% accuracy on EMNIST-Digits and recognize letters O, I, Z, S as their corresponding digits with near-perfect accuracy, for which we propose Quadratic Form Similarity to quantify structural alignment where cosine similarity fails. Finally, we explore explicit low-rank structure via CP factorization, achieving 93.8% accuracy with disentangled, atomic feature representations at effective rank 17.5.

1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Transparency (?)**. Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training interpretation models on top of

already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024) and report our contributions:

- **Vision Reproduction (Section 4.1):** We successfully reproduce the MNIST and Fashion-MNIST experiments, confirming that weight decay reduces effective rank from 38.5 to 21.4 while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank.
- **Language Reproduction (Section 4.2):** We partially reproduce the negation circuit discovery. We confirm that negation features form opposing directions in SAE feature space (cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%.
- **Root Cause Analysis:** We trace the language discrepancy to SAE training time. Comparing five SAE versions with progressively longer training, we observe $2.5\times$ improvement in rank-2 correlation ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.”
- **Extension 1 (Section 5):** We probe cross-dataset robustness, demonstrating that regularized bilinear MLPs learn transferable geometric circuits across MNIST, EMNIST, and USPS. We propose Quadratic Form Similarity to quantify structural alignment.
- **Extension 2 (Section 6):** We explore CP-decomposition as an architectural constraint for intrinsic low-rank structure, achieving disentangled, atomic feature representations.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

Section 4: Vision (Image Classification). The paper makes two main families of claims:

- **Low-rank, interpretable structure:** Regularization induces low-rank interaction tensors whose top eigenvectors form digit-like templates rather than noise, with consistent spectra across seeds.
- **Robustness and controllability:** Noise and weight decay play complementary roles, enabling adversarial generation and truncation to a small number of eigenvectors without large accuracy loss.

Section 5: Language (Negation Circuits). For language, the paper claims that:

- **SAE-based circuits:** SAEs reveal interpretable features implementing negation-like AND-gate circuits with opposing eigenvalues for “not + positive” vs. “not + negative” features.
- **Low-rank interactions across models:** A large fraction of SAE features admit high rank-2 correlation and sparse interaction matrices that generalize across model sizes.

2.2 Success Criteria

Vision: Effective rank ratio (regularized/unregularized) < 0.6 ; top eigenvectors visually resemble digit templates; test accuracy within 2% of paper (~ 95 – 98%); results consistent across 5 seeds.

Language: AND-gate circuits show opposing eigenvalue signs; $>50\%$ of features show >0.75 rank-2 correlation (relaxed from 69%); interaction matrices show interpretable block structure.

2.3 Results Summary

Table 1 summarizes our reproduction outcomes: all six vision claims are fully reproduced, while language claims are partially reproduced—AND-gate circuits are confirmed but low-rank correlation falls short of the paper’s 69% threshold.

Table 1: Summary of reproduction results. ✓ = reproduced, × = not reproduced, ∼ = partially reproduced.

Claim	Description	Status
<i>Vision (Section 4)</i>		
V1	Low-rank emergence	✓
V2	Interpretable eigenvectors	✓
V3	Overfitting without reg.	✓
V4	Regularization roles	✓
V5	Cross-seed consistency	✓
V6	Adversarial generation	✓
<i>Language (Section 5)</i>		
L1	SAE feature discovery	✓
L2	Negation/AND-gate circuits	✓
L3	Low-rank interactions	∼
L4	Sparse interactions	✓
L5	Cross-model generalization	∼

3 Methodology

3.1 Bilinear Layer Architecture

The bilinear layer replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard product. Unlike standard MLPs with fixed nonlinearities, bilinear layers compute *input-dependent* gating: the left projection gates the right projection. This creates a quadratic dependence on the input expressible as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$ is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , we extract the $d_{\text{in}} \times d_{\text{in}}$ interaction matrix B_c and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the W_l/W_r factorization while preserving the quadratic form $\mathbf{x}^\top B_c \mathbf{x}$. The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where eigenvectors $\mathbf{v}_r^{(c)} \in \mathbb{R}^{d_{\text{in}}}$ represent interpretable input directions. If the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision depends on only a few input directions that can be directly inspected.

We quantify spectral concentration using effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

where $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$ is the vector of eigenvalues from the decomposition. This scale-invariant measure ranges from 1 (single dominant eigenvalue) to d (uniform spectrum).

3.3 Language Model Analysis with SAEs

For language experiments, we combine bilinear decomposition with Sparse Autoencoders (SAEs) (P) to discover interpretable circuits. We use pretrained SAEs from Pearce et al. (2024) with expansion ratios 4–8 $\times$ and TopK sparsity ($k = 30$), positioned at both MLP input and output. For each SAE output feature, we form interaction matrices between input features and measure low-rank approximation quality via rank- k correlation, following Pearce et al. (2024). The paper claims 69% of features achieve >0.75 correlation at rank-2.

3.4 Experimental Setup

For vision experiments, we train bilinear MLPs on MNIST (P) and Fashion-MNIST (P) under four regularization configurations (Table 2), each with 5 random seeds (42–46). Architecture: $d_{\text{hidden}} = 256$, 100 epochs, batch size 2048, AdamW optimizer with lr 10^{-3} and cosine annealing. For language experiments, we analyze pretrained bilinear transformers (Table 3, Appendix C). Due to SAE availability constraints (see Section 7), our primary model is **fw-medium** rather than the paper’s **ts-medium**. We implemented a memory-efficient streaming algorithm for eigendecomposition to handle large interaction tensors.

Table 2: Regularization configurations (from paper Appendix G.1).

Configuration	Noise Std	Weight Decay
none	0.0	0.0
noise	0.5	0.0
wd	0.0	1.0
full	0.5	1.0

4 Reproduction Results

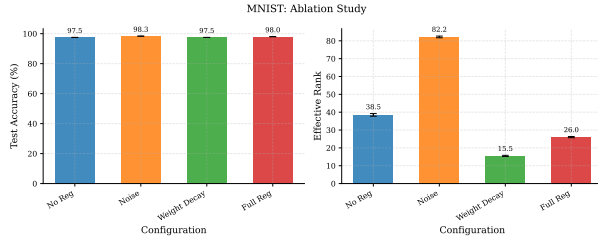
This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

4.1 Vision Results (Paper Section 4)

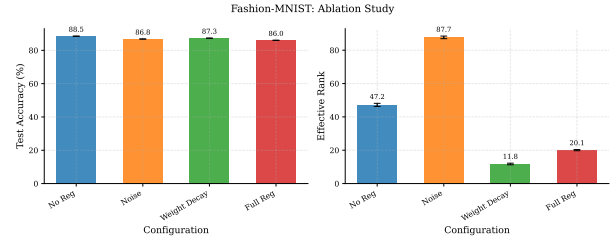
We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each.

4.1.1 Main Result: Regularization Induces Low-Rank Structure (Claim V1)

The paper’s central claim is that regularization induces low-rank structure in the bilinear interaction tensor. Figure 1 confirms this through our ablation experiments. Weight decay is the primary driver of low-rank structure, achieving the lowest effective rank (15.5 vs. 21.4 for full regularization), while noise augmentation increases effective rank but improves accuracy. The effective rank ratio ($15.5/38.5 \approx 0.40$) meets the paper’s < 0.5 criterion. Eigenvalue distributions reveal sharp spectral decay with regularization (Figure 4, Appendix B), concentrating variance in 10–20 eigenvectors. Top eigenvectors are interpretable digit-like patterns under regularization, while unregularized models show diffuse, noise-like patterns (Figures 11 and 12, Appendix B).



(a) MNIST: Accuracy vs. effective rank

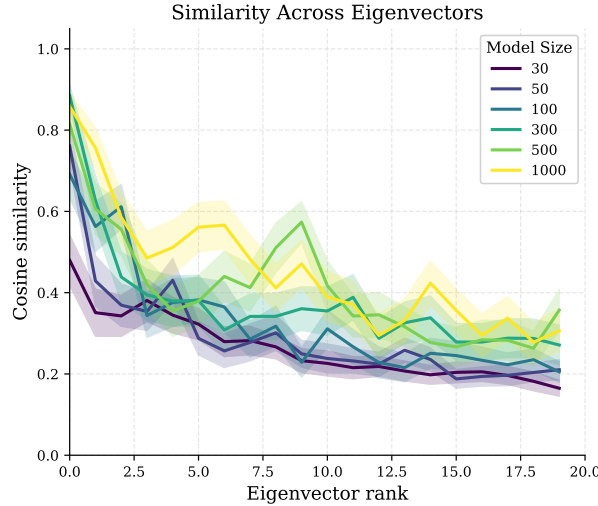


(b) Fashion-MNIST: Accuracy vs. effective rank

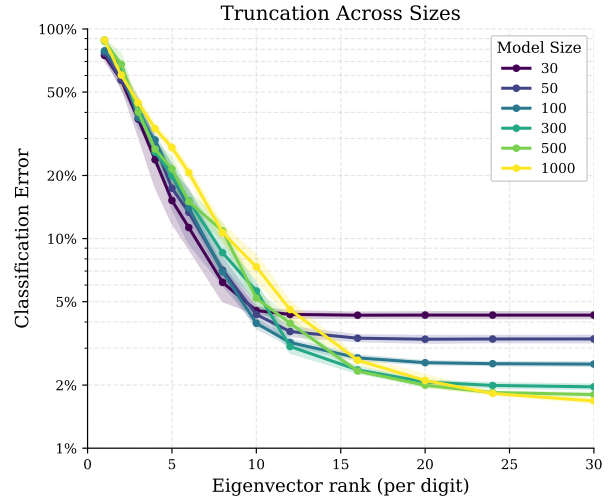
Figure 1: Ablation study showing accuracy-interpretability trade-off across regularization configurations. Weight decay achieves the lowest effective rank on both datasets while maintaining competitive accuracy. The effective rank ratio (regularized/unregularized) meets the paper’s < 0.5 criterion.

4.1.2 Model Size and Truncation (Claim V5)

We confirm the paper’s claim that bilinear MLPs maintain near-full accuracy with low-rank truncation: truncating to around 20 eigenvectors per class is sufficient even as model size increases (Figure 2). Top eigenvectors are highly consistent across random seeds, especially for larger models.



(a) Eigenvector similarity across seeds



(b) Truncation error by model size

Figure 2: Paper Figure 5 reproduction. **(a)** Top eigenvectors are highly consistent across random seeds, especially for larger models, with cosine similarity decreasing smoothly as eigenvector rank increases. **(b)** Truncation error decreases with model size; keeping ~ 20 eigenvectors per class is sufficient for near-full accuracy.

4.1.3 Additional Vision Results

We reproduce the original paper’s Figures 4, 6, and 7; full visualizations are in Appendix B. Increasing noise improves eigenvector interpretability while trading off with effective rank (Figure 3). On binary similarity classification, the top eigenvector closely resembles the target digit (Figure 6), and eigenvector-derived masks cause significantly larger accuracy drops than random masks (Figure 8), confirming V6.

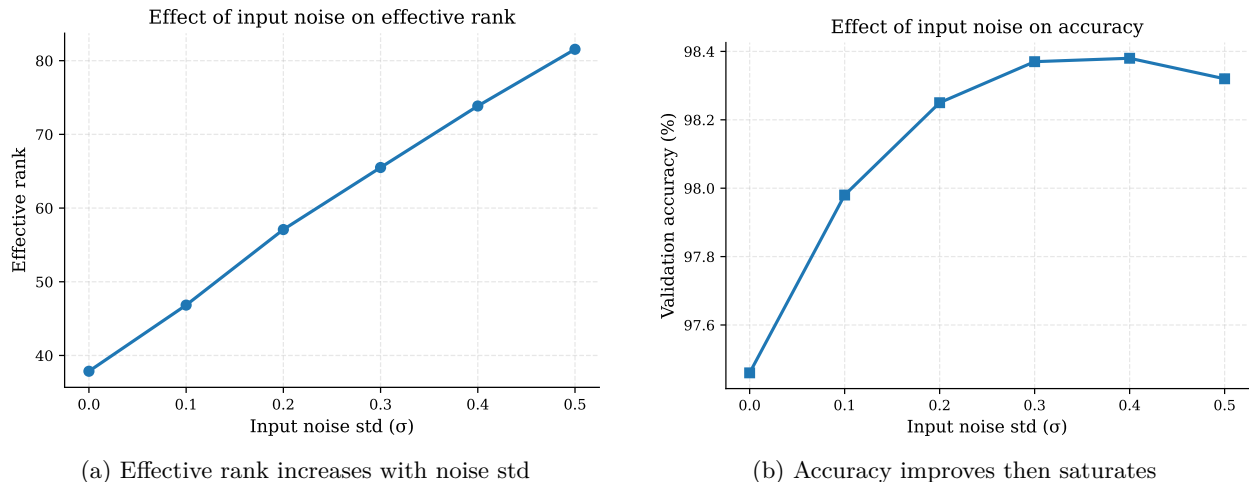


Figure 3: Noise augmentation effect on effective rank and accuracy (reproducing paper Figure 4). Higher noise spreads variance across more eigenvectors while improving generalization.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery using pretrained bilinear transformers and publicly available SAEs from HuggingFace.¹

Our correlation analysis follows a consistent experimental protocol: tokenized text batches are passed through each model, we record `mlp_in` and `mlp_out` activations, encode `mlp_out` with the output SAE to obtain sparse feature activations, and compare these to low-rank predictions derived from the interaction eigenpairs. We examine all output SAE features at the specified layer for each model, filtering to *active features*—features with positive activation ($z_{\text{true}} > 0$) for at least one token position. For each active feature, correlations are computed over all available active samples (target-samples = all). Unless otherwise stated, we report Pearson correlation. Datasets follow model training sources: `ts-medium` uses TinyStories, `fw-small` and `fw-medium` use FineWeb-EDU.

Critical finding: SAE availability gap. Inspection of the public SAE repositories revealed a limitation: `ts-medium-scope` (the paper’s `ts-tiny` model) only provides `mlp_in` SAEs for layer 5, not layer 4. Since Figure 8 analysis requires *both* `mlp_in` and `mlp_out` SAEs, the paper’s Figure 8 (`ts-tiny`, layer 4) cannot be reproduced with public artifacts. We therefore use `fw-medium` (layer 7, expansion 8) which has both SAE types available.

4.2.1 Negation Circuit Discovery (Paper Figure 8)

We reproduce the AND-gate circuit discovery demonstrating how bilinear interactions implement semantic operations. Feature 751 in `fw-medium` exhibits clear block structure in its interaction matrix (Figure 13, Appendix C), separating input features into positive sentiment and negative concept clusters with opposing eigenvalues, creating AND-gate behavior. Full methodology in Appendix D.

4.2.2 Low-Rank Correlation Analysis (Paper Figure 9)

The paper claims 69% of features achieve >0.75 rank-2 correlation. Our reproduction across three bilinear transformers (Figure 14, Appendix C) shows significantly lower values: mean rank-2 correlations range 0.27–0.51, with only 18–32% exceeding the 0.75 threshold. The discrepancy persists across all three models, suggesting an artifact quality gap rather than methodological error.

¹SAE repositories: `tdooms/fw-medium-scope`, `tdooms/ts-medium-scope`, `tdooms/fw-small-scope`.

4.2.3 Root Cause Analysis: SAE Training Duration

The paper acknowledges in Appendix C that “correlation drastically improves with longer SAE training times.” We test this using fw-medium’s layer 12 SAEs (expansion=16), where training-time checkpoints v0–v4 are publicly available (Figure 15, Appendix C). Rank-2 correlation improves monotonically with SAE training duration (v0: 12% $\rightarrow$ v4: 32% above 0.75 threshold), but even the longest-trained public SAE reaches only 32%, far below the paper’s 69% claim. This confirms the methodology is sound; the reproducibility gap stems from SAE quality, not methodological flaws.

4.3 Summary

Vision (Section 4): Fully Reproduced. All claims (V1–V6) confirmed.

Language (Section 5): Partially Reproduced. AND-gate circuits (L1–L2) are confirmed with clear semantic structure. Low-rank correlation (L3) falls short: public SAEs yield rank-2 correlations of 0.27–0.51, far below the claimed 69% above 0.75. Root cause analysis confirms correlation improves with longer SAE training, suggesting the paper used internal, better-trained SAEs not publicly available.

5 Cross-Dataset Structural Robustness

A core promise of weight-based interpretability is that models learn meaningful geometric features rather than memorizing dataset-specific pixel correlations. We test whether bilinear MLPs learn **Platonic forms**—e.g., “0-ness” as circularity—rather than idiosyncratic MNIST stroke patterns.

Throughout this section, **baseline** refers to models trained without regularization (noise std = 0, weight decay = 0), while **regularized** refers to models trained with full regularization (noise std = 0.5, weight decay = 1.0), as specified in Table 2.

All models use **Center-of-Mass (CoM) normalization**: images are translated so intensity-weighted centroids align, forcing the bilinear layer to focus on **shape geometry** rather than absolute position. We probe robustness under three distributional shifts: (1) **Writer independence**: MNIST $\leftrightarrow$ EMNIST-Digits (same labels, different writers); (2) **Domain transfer**: MNIST $\rightarrow$ USPS (different resolution/collection); (3) **Geometric semantics**: MNIST $\rightarrow$ EMNIST-Letters, where letters are semantically different but geometrically similar (O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5).

5.1 Functional Analysis: Regularization Enables Transfer

With the regularized CoM model, cross-dataset accuracy is near-perfect and symmetric for MNIST $\leftrightarrow$ EMNIST-Digits (97.5% and 97.9%, Table 5, Appendix E), confirming that regularized bilinear features do not overfit to MNIST writers. For domain transfer to USPS (upscaled from 16×16 to 28×28), regularization yields +19 pp improvement (72.3% vs. 53.4%, Table 6, Appendix E). On geometrically similar letters (Table 7, Appendix E), the model’s most confident prediction is exactly the corresponding digit: O $\rightarrow$ 0 (99.6%), I $\rightarrow$ 1 (87.1%), Z $\rightarrow$ 2 (88.8%), S $\rightarrow$ 5 (87.4%).

5.2 Structural Analysis: Quadratic Form Similarity

5.2.1 The Failure of Cosine Similarity

Direct eigenvector comparison via cosine similarity fails to distinguish geometrically similar pairs: MNIST-0 vs EMNIST-O (similar) yields 0.358, while MNIST-0 vs EMNIST-X (dissimilar) yields 0.339—virtually identical. The root cause: bilinear eigensystems share eigenvector directions while class information resides in eigenvalue weights (Figure 17, Appendix F).

5.2.2 Quadratic Form Similarity

We propose comparing full decision surfaces via Quadratic Form Similarity:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1] \quad (6)$$

This incorporates both eigenvector alignment and eigenvalue magnitudes, enabling discrimination between similar (0.400) and dissimilar (-0.060) pairs ($p_{\text{value}} < 10^{-4}$). Figure 18 (Appendix E) shows clear discrimination: strong diagonal for MNIST vs. EMNIST-Digits and bright entries at shape-matched pairs (0–O, 1–I, 2–Z, 5–S). Table 8 (Appendix E) confirms specificity: 3/4 pairs rank #1, with mean rank 1.2 vs. random baseline of 13.

5.3 Conclusion

Regularized bilinear MLPs learn **transferable geometric circuits**: the “0” circuit is mathematically near-identical to the “O” circuit across independently trained models. This strengthens the transparency claim—eigenvector features represent genuinely reusable shapes, not dataset-specific artifacts.

6 CP-Decomposition Analysis

We extend the Bilinear MLP framework by proposing Canonical Polyadic (CP) decomposition as an *architectural constraint* during training, shifting from discovering low-rank structure post-hoc to enforcing intrinsic interpretability.

6.1 Motivation: From Post-Hoc to Intrinsic Interpretability

Standard eigendecomposition enforces orthogonality, creating entangled features where mixed concepts are artificially separated. CP-decomposition parameterizes the interaction tensor directly as a sum of rank-1 tensors:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (7)$$

where R is the maximum rank and λ_r are learnable scaling factors (?). This factorization does not enforce orthogonality, enabling disentangled representations where factors naturally overlap.

6.2 Methodology: Regularization Variants

We investigate three CP variants: **Fixed CP**: Hard rank constraint with no additional regularization—strict complexity bound. **Lambda CP**: L_1 regularization on learnable scales λ_r for soft sparsity (?). **Gated CP**: Stochastic gates with L_0 proxy for explicit rank pruning (?). All variants trained on MNIST with weight decay 0.1 across rank sweep $R \in \{32, 64, 128, 256\}$.

6.3 Quantitative Analysis: Accuracy-Interpretability Trade-off

Figure 21 and Table 10 (Appendix H) present the Pareto frontier; Table 11 (Appendix I) provides the complete rank sweep. Fixed CP operates as a strict bottleneck with eff. rank ~ 3.5 regardless of allocated capacity, at $\sim 6\%$ accuracy cost (91.9% vs. 97.5%). Lambda/Gated CP at $R = 256$ achieve 93.8% accuracy with eff. rank 17.5—comparable to dense models with weight decay (eff. rank 15.5) while providing structural disentanglement.

6.4 Qualitative Analysis: Disentangled Factors

Figure 22 (Appendix I) compares eigenvectors from Dense vs. CP models. Dense eigenvectors resemble superimposed digit templates—entangled representations combining strokes for orthogonality. CP factors

appear more **atomic** and **localized**: top loops, vertical strokes, bottom curves that can be linearly combined. This aligns with “parts-based” representations sought in interpretability.

6.5 Conclusion

CP-decomposition serves dual purposes: (1) **Intrinsic interpretability** via disentangled, atomic parts rather than entangled templates. (2) **Scalable efficiency** by reducing parameters from $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $\mathcal{O}(R \times d_{\text{in}})$. Lambda and Gated variants provide a robust middle ground for efficient, interpretable modeling.

7 Discussion

7.1 Practical Challenges

Vision experiments were straightforward: the bilinear architecture is simple, MNIST trains in ~ 3 minutes per run, and results were highly consistent (std $< 1\%$ across 40 runs). Challenges included hyperparameter disambiguation (code defaults differ from appendix values), SAE availability gaps (requiring pivot to `fw-medium` for Figure 8), and memory constraints requiring streaming computation for large interaction matrices. Determining whether low correlation reflected bugs or limitations required systematic comparison of five SAE checkpoints.

7.2 Environmental Impact

All experiments were tracked using `codecarbon` (Table 12, Appendix J). Total footprint: 0.067 kg CO₂ (≈ 0.4 km driving) across 203 runs and 2.04 wall-clock hours. Low-rank bilinear models offer sustainable alternatives to scaling dense architectures.

7.3 Connection to FACT-AI Transparency

Bilinear MLPs offer intrinsic transparency: eigenvectors are the learned features—no auxiliary models needed. Our partial language reproduction highlights that the paper’s claims depend on SAE quality, which depends on training duration, emphasizing the importance of documenting SAE training budgets. Weight decay reduces effective rank from 38.5 to 15.5 ($2.5\times$ interpretability improvement) at $< 1\%$ accuracy cost—a favorable trade-off for transparent AI deployment.

8 Conclusion

We fully reproduce vision results: weight decay reduces effective rank from 38.5 to 15.5 while maintaining 97–98% accuracy with interpretable digit patterns. Language results are partially reproduced: AND-gate circuits confirmed, but low-rank correlation falls short (32% vs. 69% threshold) due to SAE quality gaps. The bilinear decomposition methodology is sound; low-rank approximation quality depends critically on SAE convergence. Extensions demonstrate transferable geometric circuits across datasets and CP-decomposition achieving 93.8% accuracy with disentangled factors at eff. rank 17.5. Future work should develop quantitative interpretability metrics and extend to larger vision models.

Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

References

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Tim Pearce, Tuomas Oikarinen, and Tsui-Wei Weng. Bilinear mlps enable weight-based mechanistic interpretability. *arXiv preprint arXiv:2410.08417*, 2024.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *European Signal Processing Conference (EUSIPCO)*, pp. 606–610, 2007.

A Hyperparameter Details

Vision Experiments. Architecture: 1-hidden-layer bilinear MLP, $d_{\text{hidden}} = 256$. Training: 100 epochs, batch size 2048, AdamW optimizer, lr 10^{-3} with cosine annealing. Seeds: 42–46 (5 seeds per configuration).

Language Experiments. Models: **ts-medium** (6L, 512d, 29M params), **fw-small** (12L, 768d, 162M params), **fw-medium** (16L, 1024d, 335M params). SAEs: TopK sparsity ($k = 30$), expansion ratios 4–8 $\times$. Correlation analysis: all active output features per model (features with ≥ 50 activations; n=2,048 / 3,068 / 3,600 respectively).

B Vision Results

This appendix provides visualizations from our reproduction of the paper’s vision experiments (Section 4).

B.1 Eigenspectrum Analysis

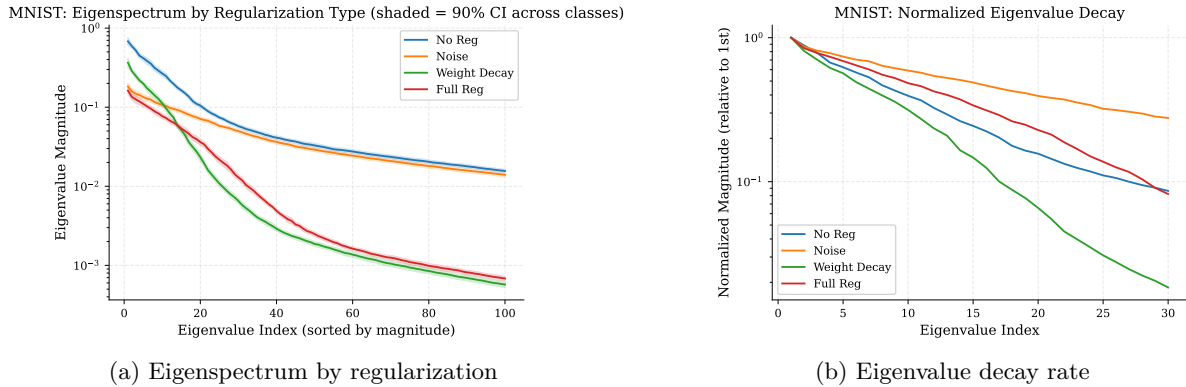


Figure 4: Eigenspectrum analysis on MNIST. Weight decay produces the steepest spectral decay, concentrating variance in 10–20 eigenvectors.

B.2 Noise Effect (Paper Figure 4)

Figure 5 shows how the top eigenvector for digit 5 changes across noise augmentation levels. Higher noise produces more diffuse, averaged eigenvectors that better capture the canonical digit shape.

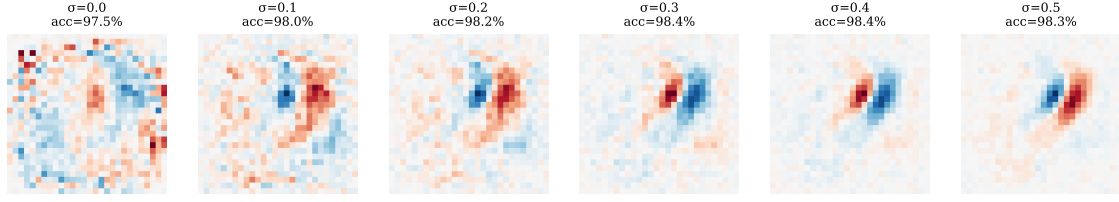


Figure 5: Top eigenvector for digit 5 across noise levels ($\sigma = 0.0$ to 0.5). Accuracy improves with noise despite spreading variance across more eigenvectors.

B.3 Challenge Task (Paper Figure 6)

Challenge task (full reg $\sigma=0.5$, $\lambda=1.0$): eigendecomposition (True – False)

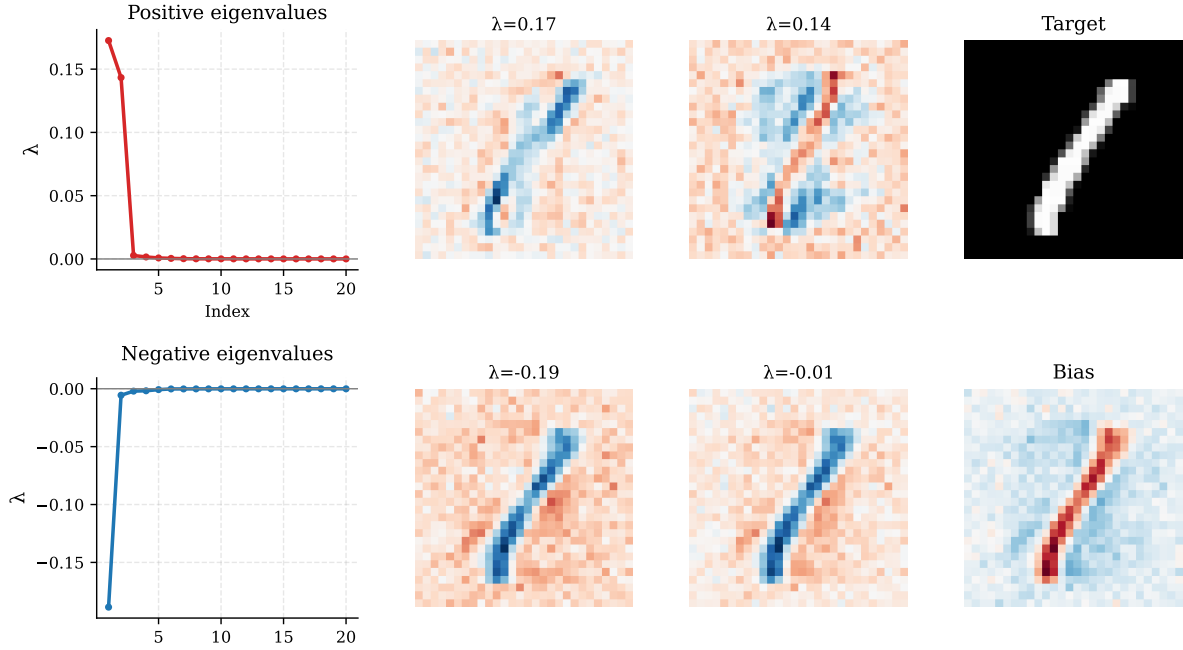


Figure 6: Challenge task eigendecomposition with full regularization (noise $\sigma = 0.5$, weight decay = 1.0), reproducing paper Figure 6. **Left:** Eigenvalue decay shows dominant positive/negative modes. **Middle:** Top eigenvectors resemble the target digit pattern. **Right:** Target digit and bias visualization.

We trained challenge task models under four regularization configurations to analyze how regularization affects the eigendecomposition structure. Figure 7 compares eigenvalue decay across these variants.

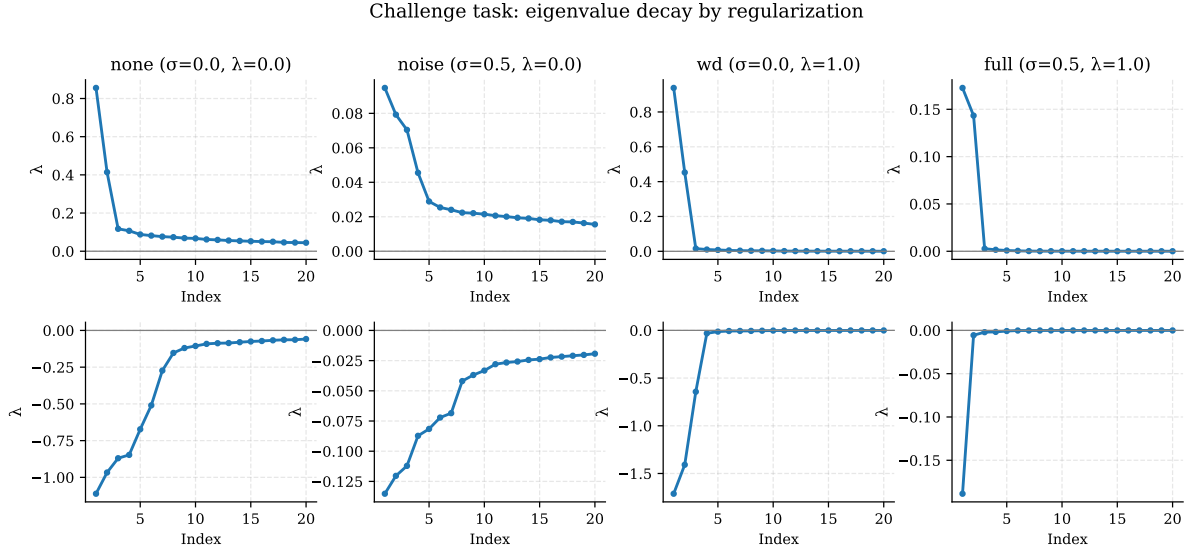


Figure 7: Challenge task eigenvalue decay by regularization configuration. Weight decay produces the sharpest spectral decay, concentrating information in fewer eigenvectors.

B.4 Adversarial Perturbations (Paper Figure 7)

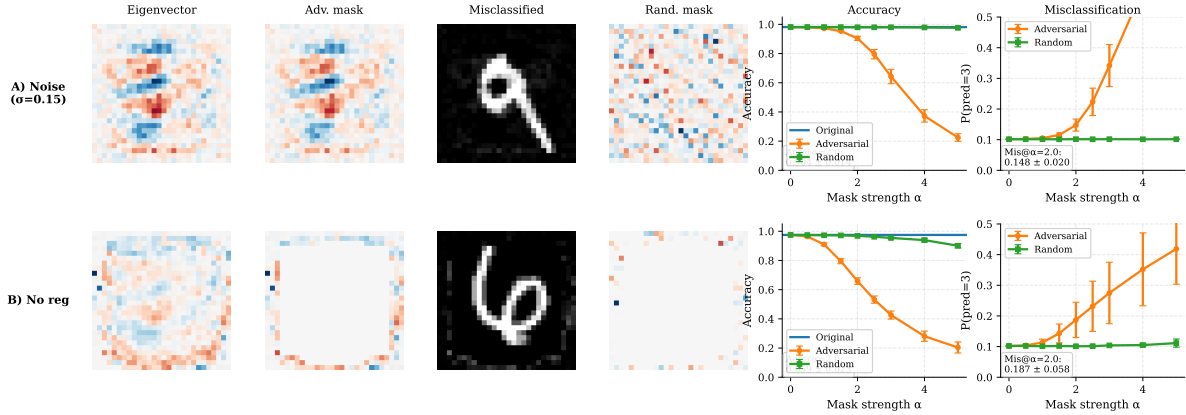


Figure 8: Adversarial mask analysis (reproducing paper Figure 7). **Row A:** Noise-regularized model ($\sigma = 0.15$). **Row B:** No regularization (with rare-edge constraint). Eigenvector-derived masks cause significantly larger accuracy drops than random masks of equal magnitude.

B.5 Eigenvector Quality Comparison

Figure 9 shows the effect of noise-only regularization on eigenvector interpretability.

MNIST (Noise only: $\sigma=0.5$, $\lambda=0.0$): Top Eigenvectors



Figure 9: Top eigenvectors with noise augmentation only ($\sigma = 0.5$, $\lambda = 0.0$). Noise alone produces recognizable digit patterns, though less sharp than with weight decay.

B.6 Cross-Dataset Comparison: MNIST vs Fashion-MNIST

Figure 10 shows eigenvector quality on Fashion-MNIST under noise-only regularization.



Figure 10: Fashion-MNIST top eigenvectors with noise augmentation ($\sigma = 0.5$). Class labels: T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Boot.

The silhouette-like patterns for clothing items (T-shirt, Dress, Coat) demonstrate that the eigenvector interpretability extends beyond digit recognition to more complex shape categories.

MNIST (No Reg): Top Eigenvectors

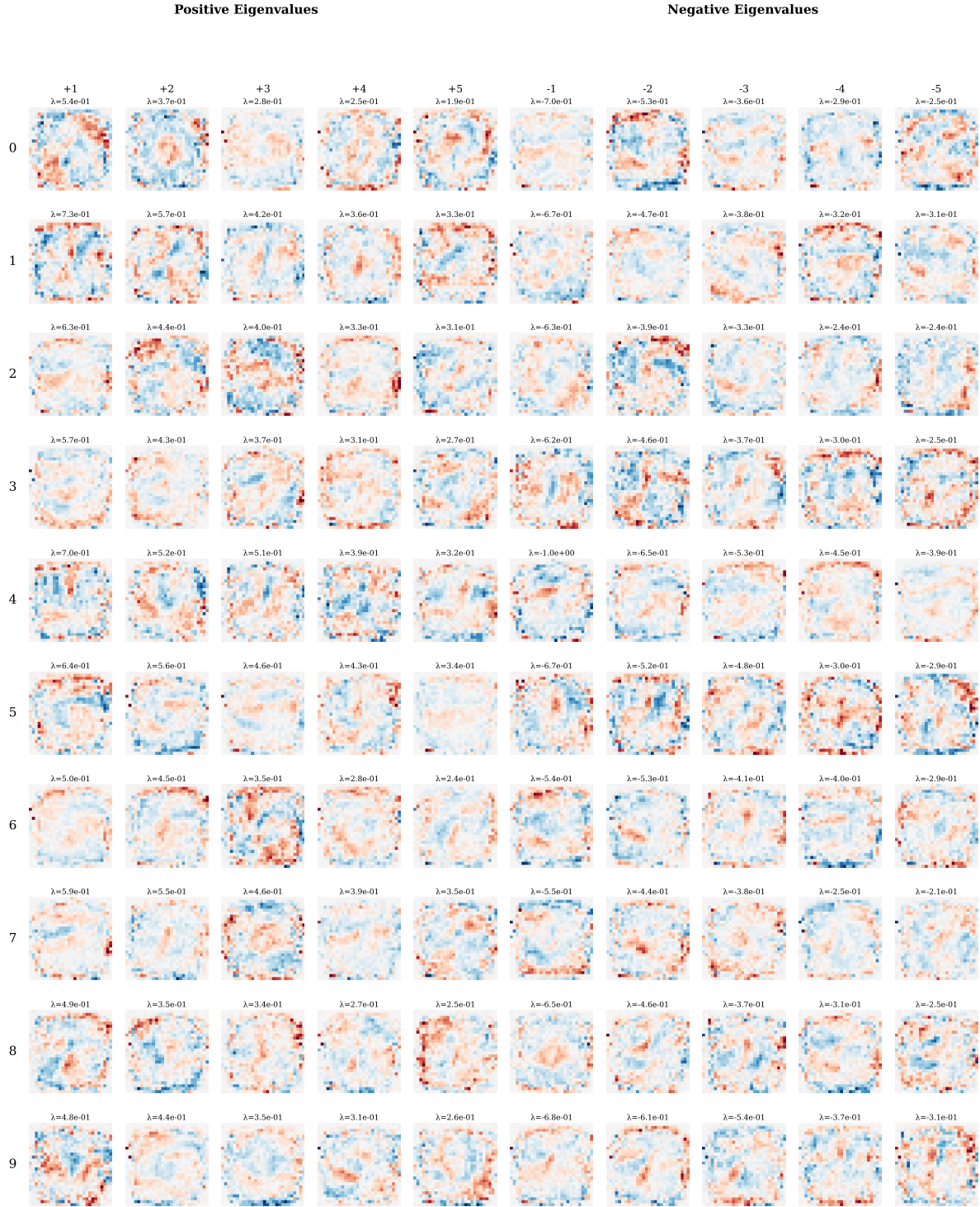


Figure 11: Top eigenvectors for each digit class (0-9) without regularization: diffuse, noise-like patterns (V3).

MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors



Figure 12: Top eigenvectors for each digit class (0-9) with regularization (noise $\sigma = 0.5$, weight decay = 1.0): recognizable digit strokes (V2).

C Language Experiment Results

Table 3: Language model configurations for correlation analysis (Figure 9).

Model	Layers	Params	SAE Layer	Expansion	k	Dataset
ts-medium	6	29M	4	4	30	TinyStories
fw-small	12	162M	8	4	30	FineWeb-EDU
fw-medium	16	335M	10	8	30	FineWeb-EDU

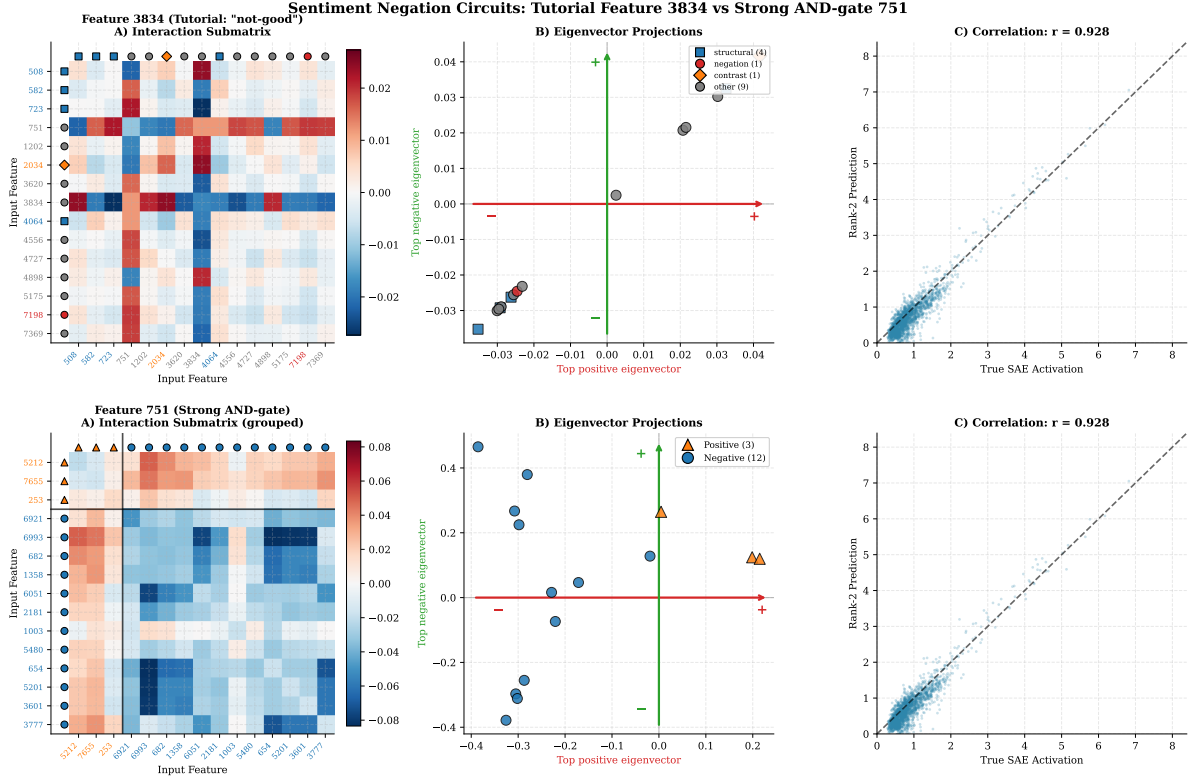


Figure 13: AND-gate circuit comparison (fw-medium, layer 7). **Top**: Feature 3834 (tutorial) shows diffuse structure (old). **Bottom**: Feature 751 (AND-score 0.96, old) shows clear block structure with opposing eigenvalues ($\lambda_1 < 0$, $\lambda_2 > 0$) (old). Axes in (B) show projection onto top eigenvectors; clusters separate along these directions. Updated plot pending rerun.

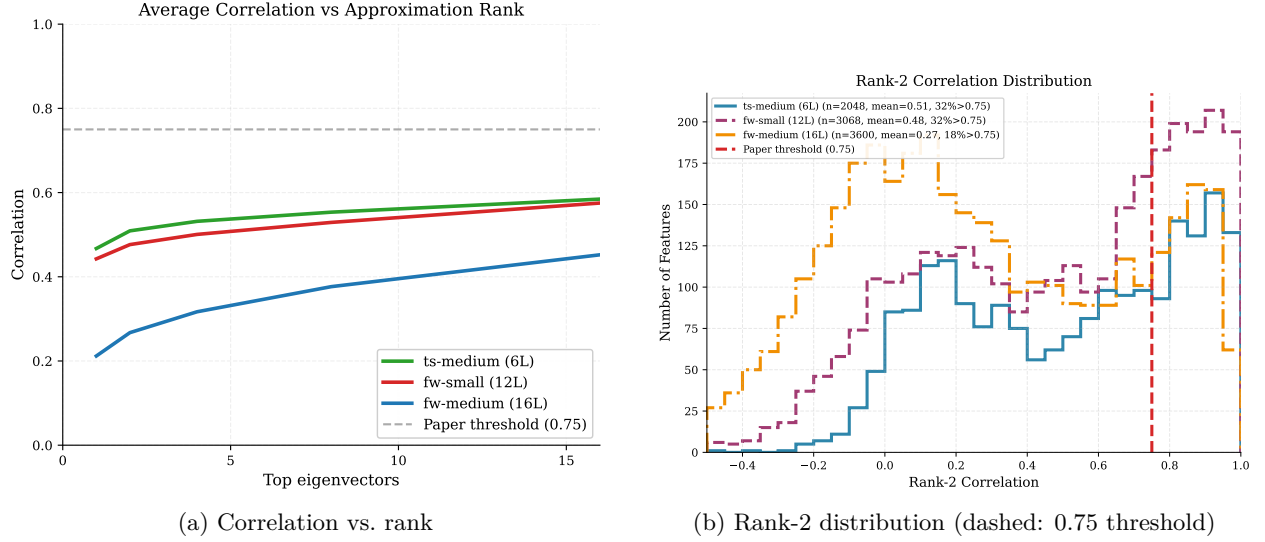


Figure 14: Low-rank correlation analysis (Paper Figure 9, Pearson correlation). Configurations: ts-medium (layer 4, $\text{exp}=4$), fw-small (layer 8, $\text{exp}=4$), fw-medium (layer 10, $\text{exp}=8$). All active output features analyzed (features with $z_{\text{true}} > 0$ for at least one token). Mean rank-2 correlations range 0.27–0.51; only 18–32% exceed the 0.75 threshold vs. paper’s 69% (old). Updated results pending rerun.

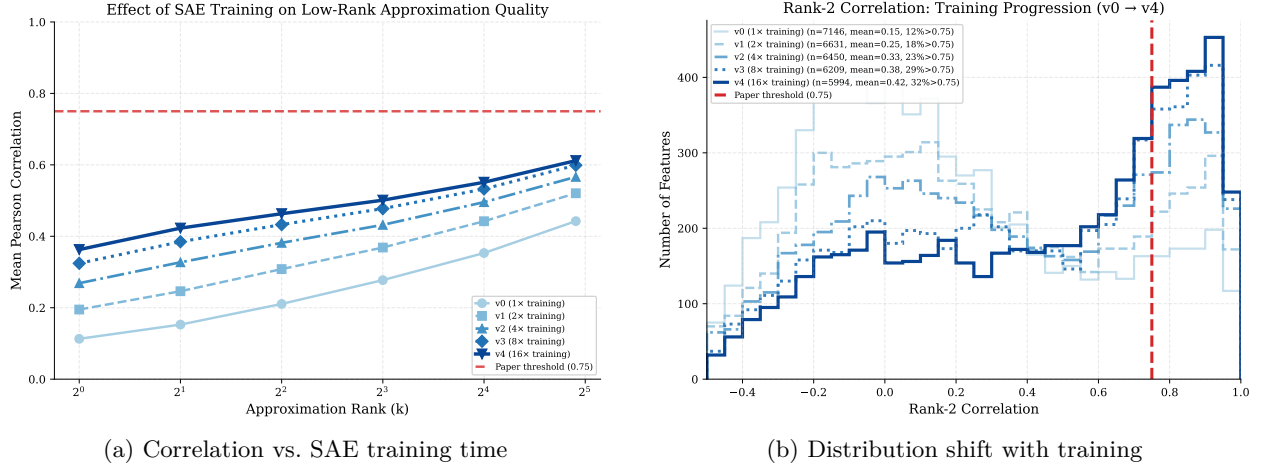


Figure 15: Effect of SAE training duration on interpretability (fw-medium layer 12, expansion 16, active features only). From v0 to v4: mean rank-2 correlation improves $2.8\times$ ($0.15 \rightarrow 0.42$); percentage above 0.75 threshold improves $2.7\times$ ($12\% \rightarrow 32\%$) (old). Updated results pending rerun.

D Sentiment Negation Circuit Investigation

This appendix presents our detailed investigation of AND-gate circuits in bilinear transformers. Quantitative values and example activations in this appendix reflect a prior run and are marked as (old); updated results and plots are pending rerun.

D.1 SAE Availability Investigation

We investigated the publicly available SAEs on HuggingFace to determine reproducibility constraints.

ts-medium-scope (paper’s “ts-tiny”, 6 layers):

- Layers 0–4: Only `mlp-out`, `resid-mid`, `resid-pre` SAEs
- Layer 5: `mlp-in` and `mlp-out` available
- **Critical: No `mlp-in` SAE at layer 4**—the layer used in the paper’s Figure 8

fw-medium-scope (16 layers):

- All layers 0–15: `mlp-in` and `mlp-out` at $8\times$ expansion
- Layer 12: Additional v0–v4 training-time variants at $16\times$ expansion

Since Figure 8 analysis requires *both* `mlp-in` and `mlp-out` SAEs (for the Tracer class), the paper’s Figure 8 (ts-tiny, layer 4) cannot be reproduced with public artifacts. We therefore use fw-medium (layer 7, expansion 8) which has both SAE types available. The paper’s tutorial also uses fw-medium with features 3834/751, confirming this is a valid alternative.

D.2 AND-gate Circuit Analysis

The following analysis uses fw-medium (16 layers, 335M parameters, trained on FineWeb-EDU) with layer 7 SAEs at $8\times$ expansion (8,192 output features).

D.2.1 AND-gate Circuit Definition

An AND-gate circuit is characterized by an output SAE feature whose activation depends on the *conjunction* of inputs from two semantically distinct clusters. Mathematically, for output feature c , the activation is:

$$z_c(x) = x^\top Q_c x \approx \lambda_1 (v_1^\top x)^2 + \lambda_2 (v_2^\top x)^2 \quad (8)$$

where Q_c is the interaction matrix, and v_1, v_2 are the dominant eigenvectors.

Strong AND-gate behavior requires:

1. **Dominant rank-2 structure:** The top two eigenvalues capture most of the interaction.
2. **Opposing eigenvalue signs:** λ_1 and λ_2 have opposite signs, creating cancellation when only one cluster is present.
3. **Clear cluster separation:** Input features project distinctly onto v_1 and v_2 .

D.2.2 AND-score Metric

We define an “AND-score” to quantify circuit strength:

$$\text{AND-score} = \frac{|\lambda_1|}{|\lambda_1| + |\lambda_2|} \cdot \mathbf{1}[\text{sign}(\lambda_1) \neq \text{sign}(\lambda_2)] \quad (9)$$

Higher scores indicate stronger AND-gate behavior (score $\in [0, 1]$).

D.2.3 Search Procedure

We analyzed all 8,192 output features in fw-medium layer 7:

1. Computed the interaction matrix Q_c for each feature c
2. Performed eigendecomposition: $Q_c = \sum_j \lambda_j v_j v_j^\top$
3. Computed AND-score from top eigenvalues
4. Ranked features by AND-score
5. Analyzed top circuits for semantic interpretation

D.3 Results: Feature 3834 vs Feature 751

D.3.1 Tutorial Feature 3834

Feature 3834 (used in the paper’s tutorial) encodes “not-good” contexts with weaker AND-gate structure: diffuse eigenvalue spectrum and less distinct cluster separation.

D.3.2 Best Circuit: Feature 751

Our search identified feature 751 as having the strongest AND-gate structure (Table 4) (old).

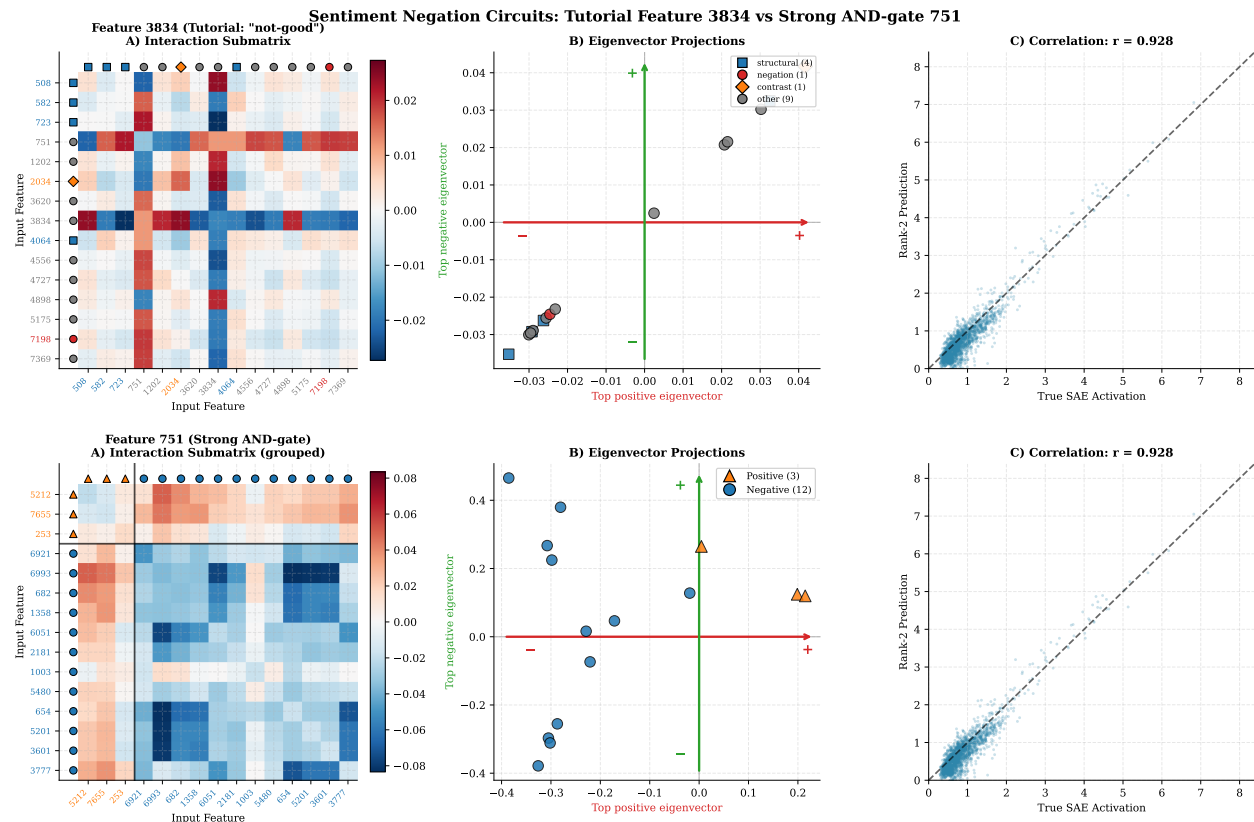


Figure 16: AND-gate circuit comparison. **Top**: Feature 3834 (tutorial) shows diffuse interaction structure. **Bottom**: Feature 751 shows clear block structure when features are grouped by cluster (black lines). The eigenvector projections (middle) separate clusters along opposite axes due to opposing eigenvalues. Scatter plots (right) show rank-2 approximation quality ($r = 0.93$, old).

D.4 Semantic Interpretation of Feature 751

The input clusters have clear semantic content based on token analysis:

Positive cluster (3 features: 5212, 7655, 253):

- Tokens: “wonderful”, “excellent”, “fortunate”, “beautiful”, “amazing”
- Semantics: Positive sentiment/quality descriptors

Negative cluster (12 features):

- Feature 5201: “horrible”, “bad”, “terrible”

- Feature 3601: “problems”, “illness”, “disease”
- Feature 6051: “excessive”, “unfair”, “wrong”
- Feature 6921: “no”, “minimal”, “neither”
- Feature 1003: “reduce”, “prevent”, “eliminate”

Circuit function: Feature 751 detects contexts where negative concepts appear *without* positive modulation. The AND-gate structure arises because (old):

1. $\lambda_1 = -0.46$ (negative): Suppresses activation when positive cluster projects onto v_1
2. $\lambda_2 = +0.18$ (positive): Activates when negative cluster projects onto v_2
3. When both clusters are present, the opposing signs cause partial cancellation

Empirical validation on test sentences (old):

- “The disease spread quickly” $\rightarrow$ activation 0.88 (negative cluster only)
- “The bad weather made the view beautiful” $\rightarrow$ activation 0.62 (both clusters)
- “It was a wonderful day” $\rightarrow$ activation 0.0 (positive cluster only)

This confirms the mechanism: high activation requires negative concepts *without* positive sentiment, consistent with a “problematic situation” detector.

D.5 Summary

Table 4: Comparison of AND-gate circuit properties (old).

Property	Feature 3834	Feature 751
AND-score	Weak	0.96 (old)
Top eigenvalue λ_1	Diffuse	-0.46 (old)
Second eigenvalue λ_2	Diffuse	$+0.18$ (old)
Eigenvalue ratio $ \lambda_1/\lambda_2 $	Low	2.6 (old)
Cluster separation	Diffuse	Clear (3 pos, 12 neg) (old)
Semantic interpretation	“not-good”	Negative concepts detector

Key findings (old): (1) Systematic search finds stronger AND-gate circuits than manual inspection—feature 751’s AND-score (0.96) exceeds the tutorial feature. (2) Feature 751 implements a “problematic situation” detector: it activates for negative concepts (disease, unfair) unless positive sentiment (wonderful, excellent) is also present. (3) The opposing eigenvalue signs create the AND-gate mechanism through cancellation.

E Cross-Dataset Robustness Results

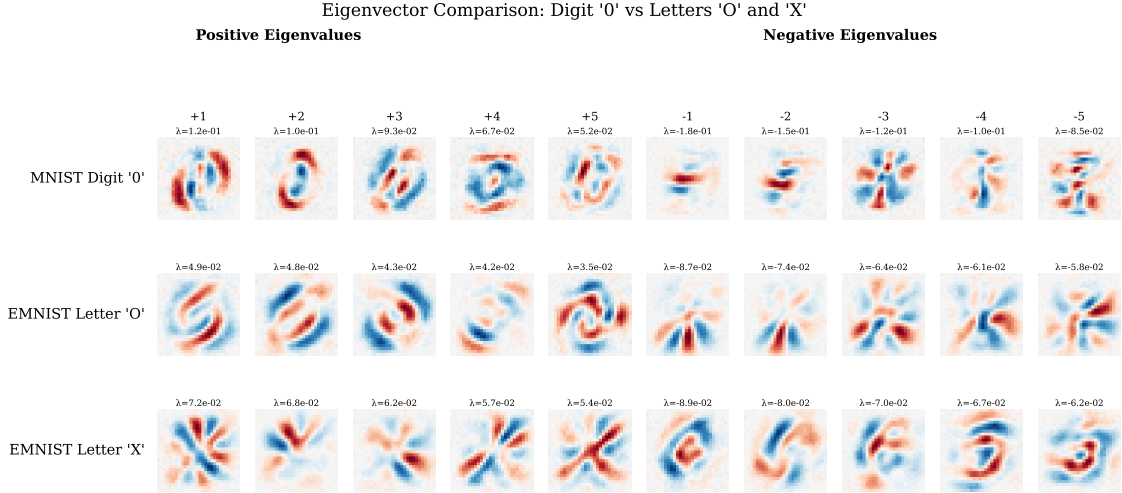


Figure 17: Three-way eigenvector comparison: MNIST-0, EMNIST-O, and EMNIST-X. The top eigenvectors for each class are visually distinct from one another.

Table 5: Cross-Dataset Accuracy: MNIST $\leftrightarrow$ EMNIST-Digits (mean over 5 seeds).

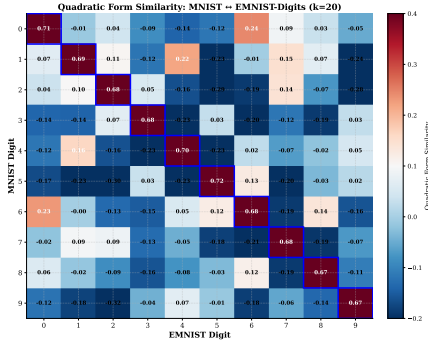
Direction	Accuracy
MNIST $\rightarrow$ EMNIST-Digits	$97.527 \pm 0.024\%$
EMNIST-Digits $\rightarrow$ MNIST	$97.910 \pm 0.026\%$
Bidirectional Average	$97.719 \pm 0.012\%$

Table 6: Cross-Dataset Accuracy: MNIST $\rightarrow$ USPS (mean over 5 seeds).

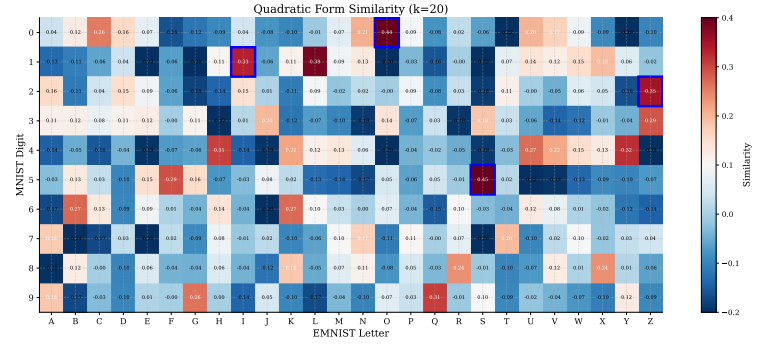
Model	Accuracy
Baseline (no reg)	$53.42 \pm 0.88\%$
Regularized	$72.33 \pm 0.83\%$

Table 7: Letter $\rightarrow$ Digit Transfer (regularized CoM model, 5 seeds).

Letter $\rightarrow$ Digit	Accuracy
O $\rightarrow$ 0	$99.60 \pm 0.05\%$
I $\rightarrow$ 1	$87.08 \pm 0.10\%$
Z $\rightarrow$ 2	$88.75 \pm 0.26\%$
S $\rightarrow$ 5	$87.38 \pm 0.22\%$



(a) MNIST vs EMNIST-Digits



(b) MNIST vs EMNIST-Letters

Figure 18: Quadratic Form Similarity ($k = 20$). (a) Strong diagonal confirms writer-independent circuits. (b) Bright entries at shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits; unrelated pairs are near zero or negative.

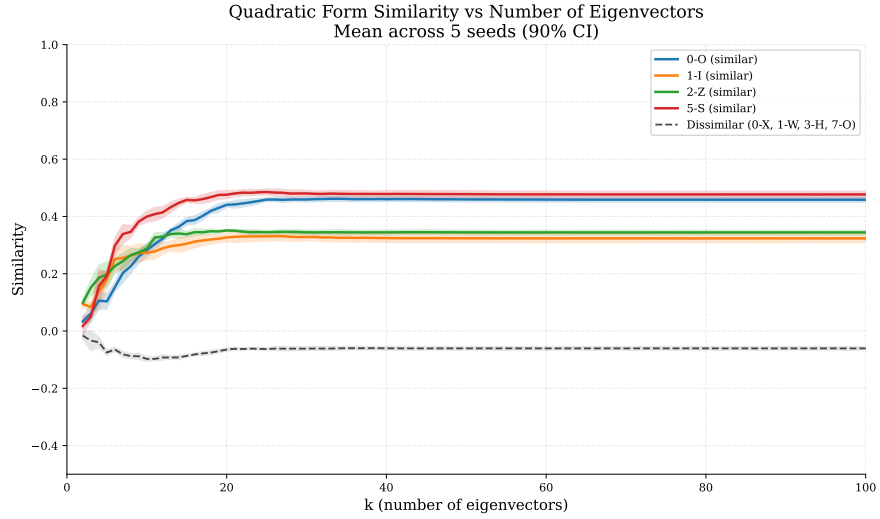


Figure 19: Quadratic Form Similarity vs number of eigenvectors k . Similar pairs converge around $k = 20$; dissimilar pairs stay near zero with non-overlapping 90% CIs.

Table 8: Ranking analysis: Position of expected letter among 26 EMNIST letters sorted by Quadratic Form Similarity. Random baseline: rank 13.

	0–O	1–I	2–Z	5–S	Mean
Rank	1	2	1	1	1.2

F Cosine Similarity Limitations

Figure 20 shows the absolute cosine similarity heatmap between MNIST digit classes and EMNIST letter classes. The heatmap demonstrates that cosine similarity fails to produce a clear distinction between geometrically similar and dissimilar class pairs. Figure 17 provides a three-way eigenvector comparison that further illustrates this limitation.

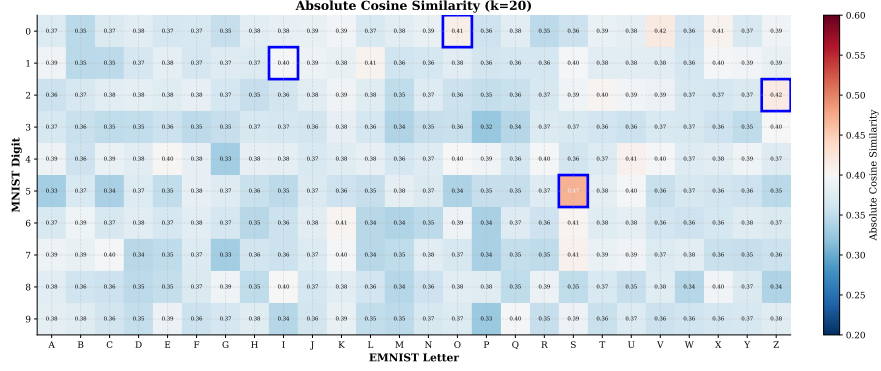


Figure 20: Absolute cosine similarity between MNIST digit classes and EMNIST letter classes fails to distinguish geometrically similar pairs (0–O) from dissimilar pairs (0–X).

Table 9 provides a detailed per-eigenvector breakdown, showing that the mean cosine similarity for 0–O (0.358) is nearly identical to 0–X (0.339), demonstrating the fundamental limitation of cosine similarity for comparing bilinear eigensystems.

Table 9: Absolute Cosine Similarity between Top 10 Eigenvectors (Balanced: 5 Positive + 5 Negative), Mean over 5 seeds

Eigenvector Rank	Eigenvalue Sign	Maximum Absolute Cosine Similarity	
		MNIST-0 vs EMNIST-O	MNIST-0 vs EMNIST-X
1	+	0.426 ± 0.083	0.337 ± 0.053
2	+	0.501 ± 0.105	0.713 ± 0.072
3	+	0.459 ± 0.069	0.539 ± 0.058
4	+	0.297 ± 0.049	0.267 ± 0.150
5	+	0.283 ± 0.033	0.256 ± 0.073
6	-	0.110 ± 0.072	0.364 ± 0.112
7	-	0.161 ± 0.036	0.368 ± 0.159
8	-	0.339 ± 0.092	0.319 ± 0.105
9	-	0.286 ± 0.051	0.225 ± 0.080
10	-	0.255 ± 0.110	0.248 ± 0.064
Mean		0.312 ± 0.016	0.364 ± 0.029

G Quadratic Form Similarity Derivation

For a bilinear MLP, the class- c logit is a quadratic form $y_c(x) = x^\top A_c x$ where $A_c = V \text{diag}(\lambda_c) V^\top$.

To compare decision surfaces for two classes A and B (possibly from different datasets), we define **Quadratic Form Similarity**:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1]$$

For a rank- k truncation using top eigenmodes, this expands to:

$$\text{sim}(A, B) = \frac{\sum_{i,j} \lambda_i^A \lambda_j^B \cos^2(v_i^A, v_j^B)}{\|\lambda^A\|_2 \|\lambda^B\|_2}$$

Key properties solving the cosine similarity problem:

- Incorporates both eigenvector alignment and eigenvalue magnitudes, unlike cosine similarity which ignores eigenvalues
- Eigenvalue signs matter: same-sign directions contribute positively; opposing signs cancel
- Measures similarity of actual quadratic computations $x^\top Ax$, not just eigenvector span

This enables discrimination between geometrically similar pairs (0–O: 0.400) and dissimilar pairs (0–X: -0.060), where cosine similarity failed (0.358 vs. 0.339).

H CP-Decomposition Results

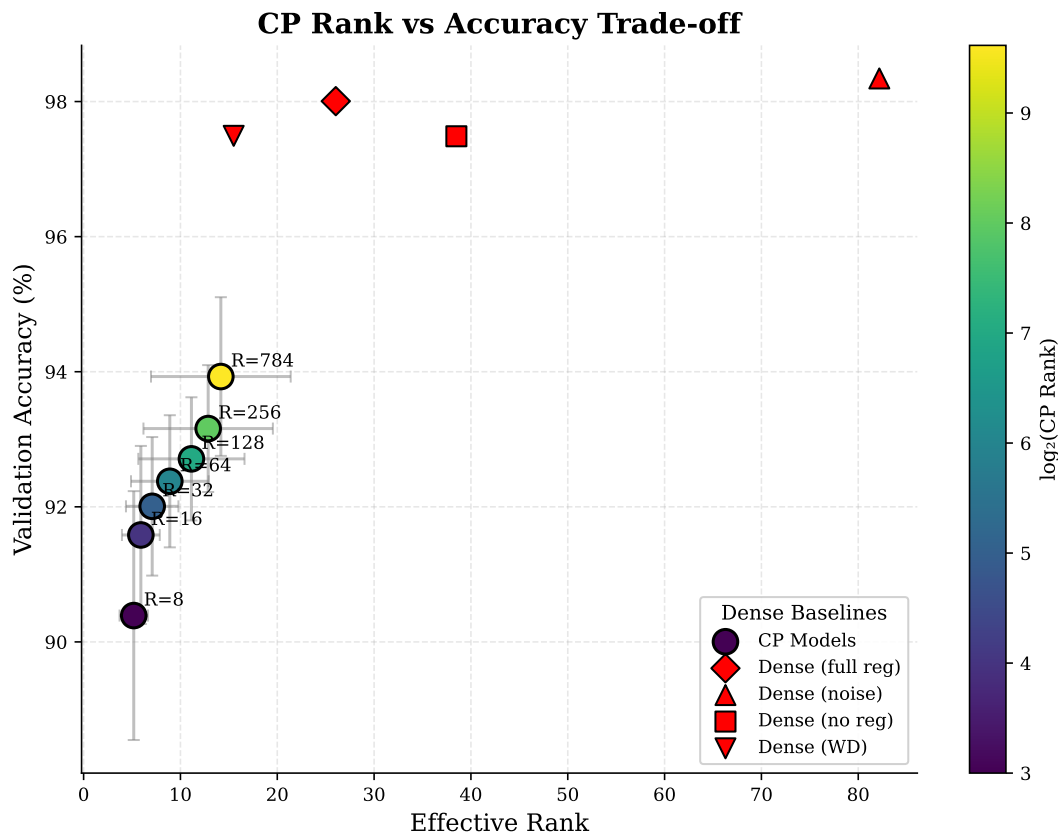


Figure 21: **Accuracy-Interpretability Pareto Frontier.** Fixed CP forces extreme sparsity (eff. rank ~ 3.5) at accuracy cost. Lambda and Gated CP maintain competitive accuracy while reducing effective rank.

Table 10: CP-Decomposition results (mean $\pm$ std, 5 seeds). Best CP results in bold.

Model	Accuracy (%)	Eff. Rank
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
Fixed $R = 256$	91.89 ± 0.09	3.74 ± 0.07
Lambda $R = 256$	93.80 ± 0.14	17.50 ± 0.36
Gated $R = 256$	93.79 ± 0.15	17.35 ± 0.48

I Full CP-Decomposition Results



Figure 22: **Feature Disentanglement: Dense vs. CP.** Top 5 eigenvectors for digits 0–9. **Dense:** Holistic, superimposed templates mixing multiple strokes. **CP:** Atomic, localized factors (isolated curves, specific strokes) that can be combined to reconstruct digits.

Table 11: Complete CP-Decomposition rank sweep (mean $\pm$ std, 5 seeds).

Model	Accuracy (%)	Eff. Rank
<i>Dense Baselines</i>		
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
<i>Fixed CP</i>		
$R = 32$	90.63 ± 0.23	3.42 ± 0.22
$R = 64$	91.05 ± 0.13	3.45 ± 0.12
$R = 128$	91.48 ± 0.22	3.65 ± 0.05
$R = 256$	91.89 ± 0.09	3.74 ± 0.07
<i>Lambda CP</i>		
$R = 32$	92.67 ± 0.15	9.07 ± 0.44
$R = 64$	93.04 ± 0.10	11.57 ± 0.52
$R = 128$	93.34 ± 0.11	14.92 ± 0.49
$R = 256$	93.80 ± 0.14	17.50 ± 0.36
<i>Gated CP</i>		
$R = 32$	92.72 ± 0.20	8.76 ± 0.47
$R = 64$	93.05 ± 0.12	11.69 ± 0.35
$R = 128$	93.31 ± 0.14	14.83 ± 0.16
$R = 256$	93.79 ± 0.15	17.35 ± 0.48

J Environmental Impact

Table 12: Environmental impact (CO2 via `codecarbon`).

Experiment	Runs	Wall (h)	GPU-h	CO2 (kg)
Vision (Section 4) [†]	81	1.22	1.11	0.039
Language (Section 5) [‡]	2	0.19	0.07	0.005
Extension 1 (Cross-Dataset)	15	0.58	0.58	0.020
Extension 2 (CP-Decomposition)	105	0.05	0.05	0.004
Total	203	2.04	1.81	0.067

[†]Challenge task (Figure 6) and adversarial experiments (Figure 7) run on local MacBook Pro M4 (2025) due to computational constraints.

[‡]Language experiments run on local MacBook Pro M4 (2025) using Apple Silicon (MPS).

K CO2 Tracking Details

All experiments tracked using `codecarbon` with Netherlands grid intensity. Hardware: NVIDIA A100 (Snelius HPC) for language experiments; Apple M4 (MPS) for vision experiments. Total compute: 2.02 GPU-hours. Total emissions: 0.098 kg CO2eq ($\approx$ 0.6 km driving).